Predicting Coronary Heart Disease from Survey Data: Data Cleaning Matters More than the Model

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Abstract

We predict coronary heart disease (MICHD) from 321 self-reported features of the 2015 BRFSS survey, covering 328,135 respondents of whom only 8.8% are positive. Using only NumPy, we compare the six linear methods of the course on one shared preprocessing pipeline and evaluate them with 5-fold cross-validated F1. Treating the survey's "don't know/refused" codes as missing values and one-hot encoding categorical answers gives the largest improvement (+0.017 F1), whereas regularization strength, squared features and class weighting change nothing beyond noise. Our best linear model, regularized logistic regression with an F1-tuned threshold, reaches a CV F1 of 0.428 ± 0.003 and a test F1 of 0.440. [TODO: one sentence on the SVM and the final model]

1 Introduction

Cardiovascular diseases are among the leading causes of death worldwide, and early identification of people at risk can support prevention. We predict whether a respondent of the Behavioral Risk Factor Surveillance System (BRFSS) reports coronary heart disease from their lifestyle and health answers, as a binary classification problem.

The data is hard for three reasons. Many questions are asked only to some respondents, so most columns are largely empty. Answers such as "don't know" or "refused" are stored as numbers (7, 9, 77, 99, ...), which a linear model reads as real amounts. Finally, only 8.8% of respondents are positive, so accuracy is misleading and we optimize F1 instead.

Our contributions are: (i) a preprocessing pipeline that handles survey codes, missing values and categorical answers using only training statistics; (ii) a comparison of the six required linear methods on identical data and identical cross-validation folds; and (iii) an ablation study showing which steps improve F1 and which do not. [TODO: add the SVM contribution if it is the final model]

2 Models and Methods

2.1 Data and preprocessing

The training set has 328,135 respondents and the test set 109,379, each with 321 features. Labels in $\{-1, 1\}$ are mapped to $\{0, 1\}$. All statistics below are computed on the training set and applied unchanged to the test set.

1. **Survey codes.** If a column's training maximum is 9, the values 7 and 9 become missing; if 99, then 77 and 99 become missing and 88 ("none") becomes 0; if 999, then 777 and 999 become missing and 888 becomes 0; if 9999, then 7777 and 9999 become missing.
2. **Column removal.** We drop 9 survey-administration columns (state, interview date, sequence numbers) and every column with more than 80% missing values, keeping 195.
3. **One-hot encoding.** The 50 columns with 3 to 10 distinct values are treated as categorical. Each becomes one indicator per value plus one "missing" indicator, giving 277 binary features.
4. **Numeric features.** The remaining 145 columns are imputed with the training median and standardized to zero mean and unit variance.
5. **Bias.** A constant column is added, for 423 features in total.

2.2 Models

We use the six methods of the course: least squares, ridge regression ($\lambda = 10^{-4}$), linear regression with the MSE loss by gradient descent (GD) and by stochastic gradient descent (SGD, batch size 1), logistic regression, and L_2-regularized logistic regression ($\lambda = 10^{-3}$, $\gamma = 0.5$, 500 iterations, zero initialization). Regression models output a score; logistic models output a probability through the sigmoid σ.

To test whether the class imbalance should be handled in the loss, we also implement a weighted logistic regression in which positive examples get weight $w_+ \in \{3, 10\}$ and negatives weight 1. Its gradient is

$$\nabla L(w) = \frac{1}{\sum_n c_n} X^\top \big(c \odot (\sigma(Xw) - y)\big) + 2\lambda w, \quad (1)$$

with $c_n = w_+$ if $y_n = 1$ and $c_n = 1$ otherwise.

[TODO: describe the L_2-regularized SVM: loss, λ, step size, iterations, label convention, and any other models]

2.3 Evaluation protocol

Models are compared with 5-fold cross-validation (random seed 1, the same folds for every model), reporting the mean ± standard deviation of F1. In each fold the decision threshold is chosen on the validation fold by searching 0.05 to 0.60 in steps of 0.01 (up to 0.95 for weighted models). The ablation steps use a single 80/20 split with the same seed. The final model is retrained on all training data and applied to the test set with the selected threshold.

3 Results

Table 1 shows the ablation: each row adds one change to the row above. The two preprocessing steps give the largest gains (+0.013 and +0.004), while tuning λ and adding squared features give none.

Table 1: Ablation on a single 80/20 split (validation F1), with test F1 where submitted.

Step	Change	Val. F1	Test F1
1	Ridge, basic cleaning	0.4106	
2	Reg. logistic, not ridge	0.4136	
3	+ survey codes as missing	0.4269	
4	+ one-hot encoding	0.4308	
5	λ tuned, best 10^{-3}	0.4315	0.439
6	+ squared numeric features	0.4315	

Table 2 compares all methods on the final preprocessing with 5-fold CV. Logistic and regularized logistic regression are best and equal; ridge is about 0.009 lower, roughly three standard deviations. The selected threshold is stable across folds (0.19–0.21 for unweighted logistic), and both submitted models score about 0.01 higher on the test set than in CV.

Table 2: Method comparison, 5-fold CV F1 (mean ± std).

Method	CV F1	Test F1
Logistic regression	0.4285 ± 0.0029	
Weighted logistic, $w_+ = 3$	0.4287 ± 0.0033	0.440
Reg. logistic, $\lambda = 10^{-3}$	0.4281 ± 0.0035	
Weighted logistic, $w_+ = 10$	0.4276 ± 0.0030	
Ridge, $\lambda = 10^{-4}$	0.4197 ± 0.0033	
MSE GD	[TODO: rerun]	
MSE SGD	[TODO: rerun]	
Least squares	singular	
L_2 SVM (teammate)	[TODO: CV]	0.446
[TODO: other models]		

4 Discussion

Preprocessing drives performance. Survey codes such as 99 ("refused") are far larger than real answers, so a linear model gives them a large, meaningless effect; turning them into missing values gave the largest single gain. One-hot encoding helps for the same reason: categorical answers stop being treated as amounts, and "don't know" becomes its own indicator.

Regularization does not matter here. With 328,135 examples and 423 features the model is far from overfitting, so λ between 10^{-5} and 10^{-3} gives the same F1, and unregularized logistic regression matches the regularized one. Squared features did not help either, likely because many numeric columns are already grouped (e.g. 5-year age bands).

Threshold tuning already handles the imbalance. Class weighting shifts the predicted probabilities upward, and the selected threshold moves with them (0.21, 0.44 and 0.52 for $w_+ = 1, 3, 10$), leaving F1 unchanged.

Method notes. Least squares cannot be solved on this feature matrix: the indicators of each categorical variable sum to the bias column, so $X^\top X$ is singular; the λI term of ridge makes it invertible. MSE gradient descent diverges with $\gamma = 0.1$, and SGD with batch size 1 needs far more than 10^5 steps to converge.

Limitations. The survey-code rule is a heuristic based on each column's maximum and misses some cases (e.g. 8 = "never" in a few questions). The ablation uses a single split, whose noise (about 0.003) is close to some of the differences. All our models are linear. [TODO: discuss whether the SVM's hinge loss explains its higher test score]

5 Summary

On the BRFSS heart-disease task, careful preprocessing (survey codes as missing, one-hot encoding, median imputation, standardization) raised F1 from 0.411 to 0.431, while model settings and class weighting gave no further gain. Logistic regression with an F1-tuned threshold is our best linear model, at CV F1 0.428 ± 0.003 and test F1 0.440. [TODO: final model, its score, and the `run.py` reproducibility statement]

References

[1] Centers for Disease Control and Prevention, "Behavioral Risk Factor Surveillance System, 2015 annual data," `https://www.cdc.gov/brfss/annual_data/annual_2015.html`.